ADITYA KUMAR SINGH

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EDUCATION

Madhav Institute of Technology & Science — Gwalior, India
B. Tech in Artificial Intelligence and Machine Learning — *Aug. 2023 – Present*

EXPERIENCE

AI Development Intern | Zipbolt Innovations — *Sep. 2025 – Jan. 2026*

- Worked on **ZipCRM**, an internal CRM platform built using the **PERN stack (PostgreSQL, Express, React, Node.js)**, contributing to backend APIs, database design, and workflow automation.
- Developed a **full-stack EV rental management system**, handling backend services, business logic, and database schema while integrating AI-powered decision workflows.
- Implemented **event-driven backend and AI orchestration using Inngest**, enabling scalable background jobs, async task execution, and intelligent automation inside the CRM.

PROJECTS

Legal RAG Assistant

- **Tech Stack:** Python, LangChain, FAISS, OpenAI Embeddings, PyMuPDF, Streamlit, AutoGen
- Built a Legal RAG Assistant using LangChain, FAISS, and OpenAI Embeddings.
- Implemented PDF processing, semantic search, and context-aware question answering.
- Developed a Streamlit interface with multi-agent (AutoGen) workflow support.

Decoder-Only Transformer (GPT-style LLM) from Scratch

- **Tech Stack:** Python, PyTorch, NumPy, Transformers (for Tokenizers), Jupyter Notebook
- Built a GPT-style Decoder-Only Transformer from scratch using PyTorch.
- Implemented multi-head self-attention, positional encoding, and a complete training pipeline.
- Developed a text generation model with tokenization, training, and inference.

Multimodal Medical Image Fusion for Brain Tumor Detection

- **Tech Stack:** Python, OpenCV, TensorFlow/Keras, VGG-19, NumPy, Flask, PyWavelets, Jupyter Notebook
- Developed a CT–MRI brain image fusion system using DWT and VGG-19.
- Implemented image registration and tumor segmentation using OpenCV.
- Built a Flask-based application for automated medical image processing.

ACHIEVEMENT

Ghummakad – AI-powered Travel Assistant (Team Evolutio – 6th place, Code Coalescence University Hackathon)

- Built an AI-powered assistant recommending affordable Indian alternatives to expensive foreign destinations using cosine similarity and ML-based recommendations.
- Integrated real-time safety alerts, eco-friendly suggestions, and revenue tools for travel businesses.
- Achieved complete MVP in 12 hours, collaborating with 5 team members under high-pressure hackathon constraints.

TECHNICAL SKILLS

Programming: Python, C++, JavaScript, SQL, R

AI/ML & Data Science: TensorFlow, PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, NLP, Deep Learning

GenAI: LangChain, crewAI, Autogen, Ollama, n8n

Research & Analytics: Market Research, Data Analysis (Excel, Google Sheets, PowerBI), Statistics & Visualization (regression, charts/graphs)

CERTIFICATIONS

- Complete Data Science, Machine Learning, DL, NLP Bootcamp
- AI in Social Media - SIP, MITS Gwalior -Dec 2024

HOBBIES

Content Creation | Connect With Aadi *Jan. 2025 – Present*

- Built a personal brand around AI, technology, and entrepreneurship by creating educational content for Instagram and YouTube.
- Produced AI-focused tutorials and short-form content, simplifying complex concepts while applying data-driven content strategies to improve audience engagement.
- Strengthened expertise in AI research, digital marketing, storytelling, and community engagement through practical, real-world content creation.